

EEG Artifact Simulation:

An open-source implementation for eye movements, power line noise, and drifts

Master Thesis Final Presentation - Maanik Marathe

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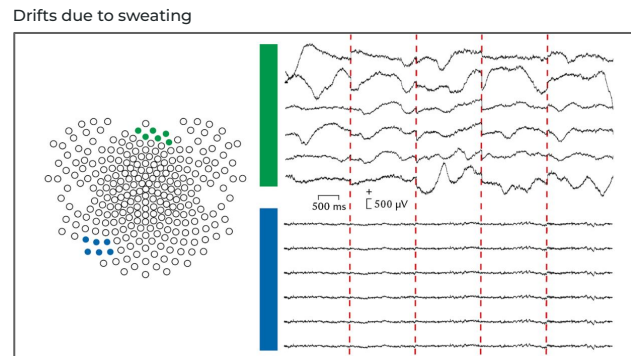
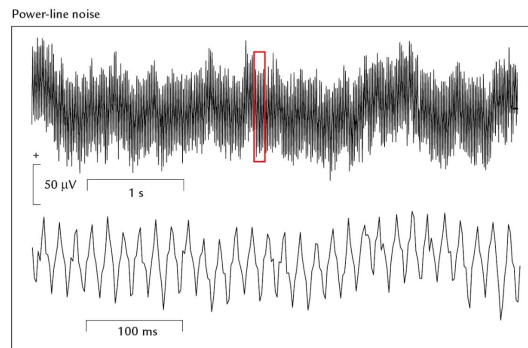
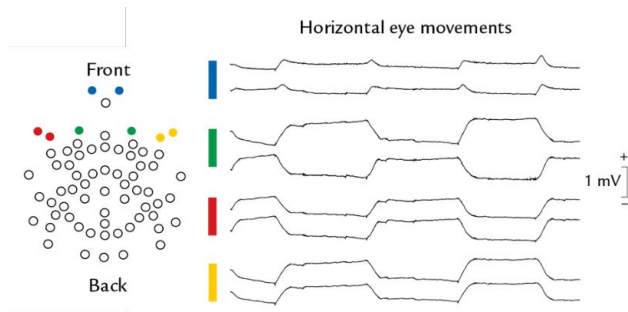
Course: M.Sc. Information Technology, Faculty 5

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Motivation

- Artifacts: undesirable signals in EEG recordings
 - From non-brain sources
 - Larger than brain-activity signal, obscures it

Q1: What do these artifacts look like?



Eye movement, power line noise, and sweating artifacts. Source: Hari, R., & Puce, A. (2017)

Motivation

- Scripts & toolboxes to detect/remove artifacts
- How to test them?
 - Use a test dataset
- Desirable properties of test dataset:
 - **Large**: sufficient instances of artifact
 - **Well-labeled**: known ground truth
 - **Suitable modality** available: e.g. eye tracking, for blink artifact processing
 - **Variation** in artifact: artifact occurrences differ from each other

Motivation - Why simulate?

- Difficult to get all these in someone else's dataset!
- ✗ Record new dataset: difficult & expensive (time, money, resources)
- ✓ Simulation:
 - Dataset size not limited
 - Control over ground truth - eg. occurrence, variation

Q2: How are these artifacts usually simulated?

Related Work - Literature

- Extract artifact from data; add to clean(ed) or simulated data

Problems:

- Low variation in artifact occurrences
- Less control over specifics of ground truth
- Custom scripts
- Often: only blink artifacts

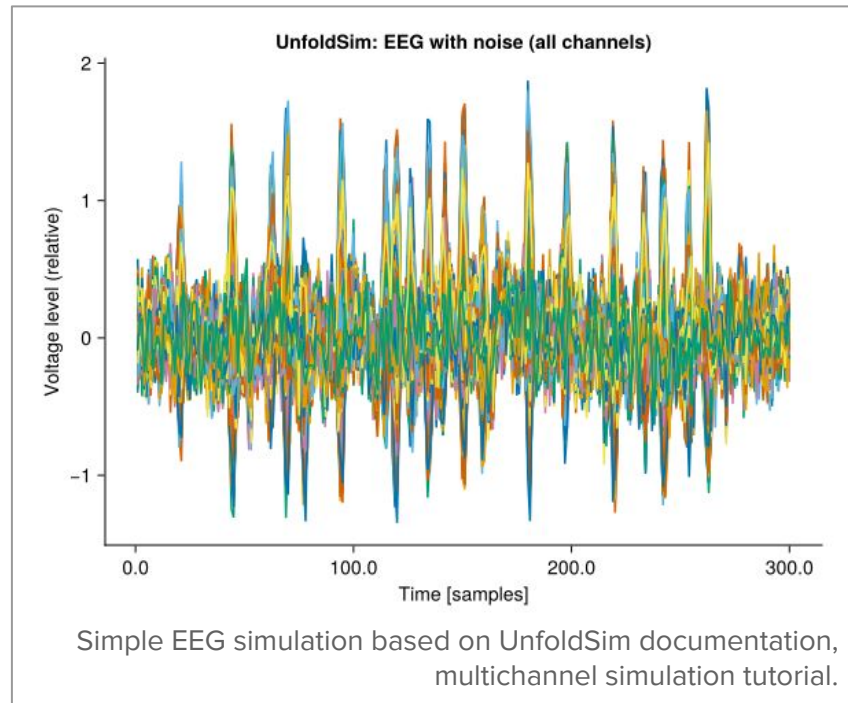
Simulation - Desired criteria

- Standardized, reproducible
 - Avoid one-off custom scripts
 - Reduce reliance on researcher's on-the-fly decisions
- Flexible
 - Easy to change simulation parameters
 - Integrate into existing code
- Open-source
 - Current approaches: often in proprietary software e.g. MATLAB
- Realistic

Q3: How can we simulate artifacts, trying to satisfy these criteria?

Reference - EEG Simulation

- Julia: UnfoldSim.jl
 - Simulation for EEG (brain) data
- Input: design, component, onset, noise
- Output: simulated EEG + random noise



Research Questions

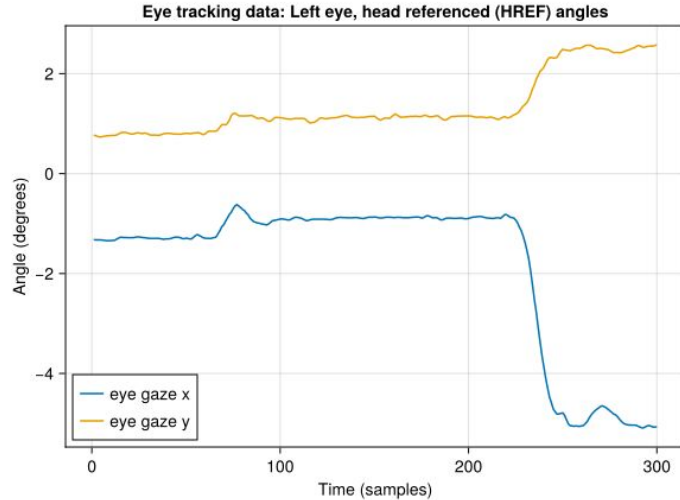
1. Given a set of chosen EEG artifacts (eye movement, power line noise and electrode drift), what is known about their origin and characteristics?
2. How are these artifacts usually simulated?
3. How can we simulate them in a way that is standardized and compatible with existing open-source EEG simulation packages?

Goals & Approach

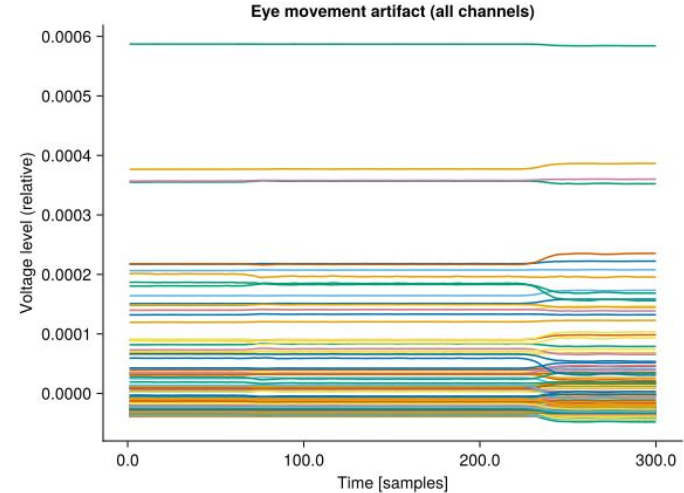
- A. Eye movement trajectory simulation:
 - a. Describe eye movement via gaze direction
 - b. Eye-models used from previous research project (Marathe, 2025)
- B. Power line noise & drifts:
 - a. Power line noise: simulate sine waves (base frequency + harmonics)
 - b. Drift: DC, linear drift, autoregressive
- C. Code Interfaces:
 - a. Integrate with UnfoldSim.jl
 - b. Flexible: Abstract types + concrete implementation
 - c. Various user-interfaces to describe artifact

Results: Eye Movement simulation

```
UnfoldSim.EyeMovement(UnfoldSim.HREFCoordinates(href_trajectory),  
                      eyemodel, "crd")
```



Real-data eye movement, left eye gaze x and y traces.
Data: Gert et al (2022)

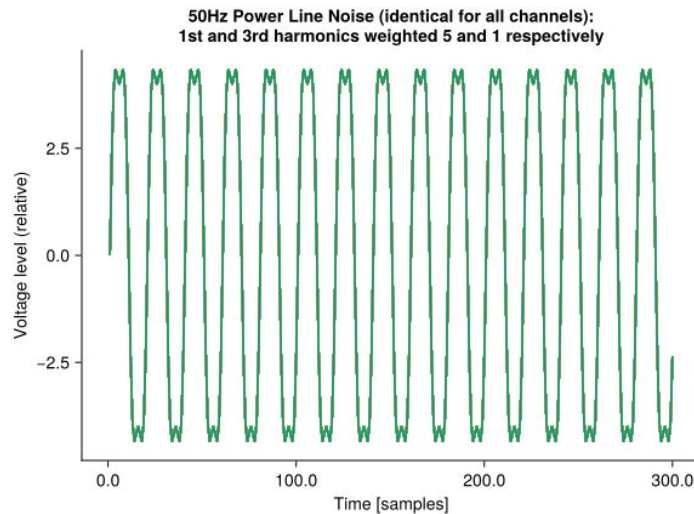


Simulated eye-movement artifact based
on the shown data.

Results: Power line noise

```
UnfoldSim.PowerLineNoise(base_freq=50, harmonics=[1 3],  
                           weights_harmonics=[5 1], sampling_rate=1000)
```

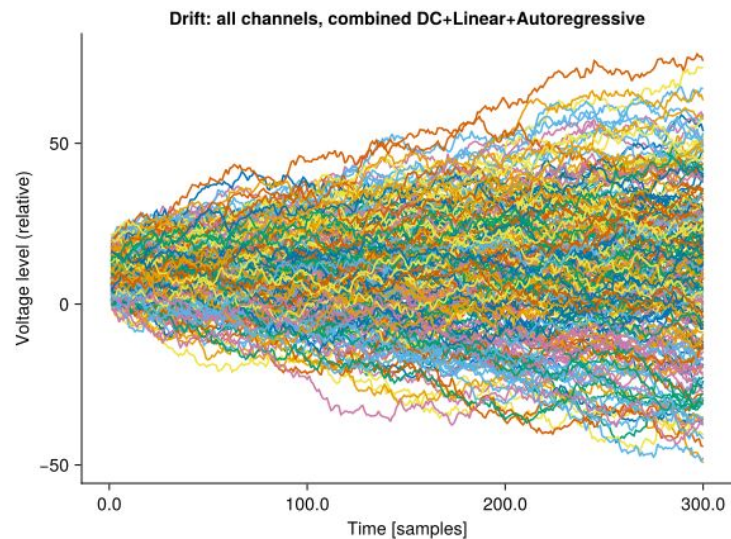
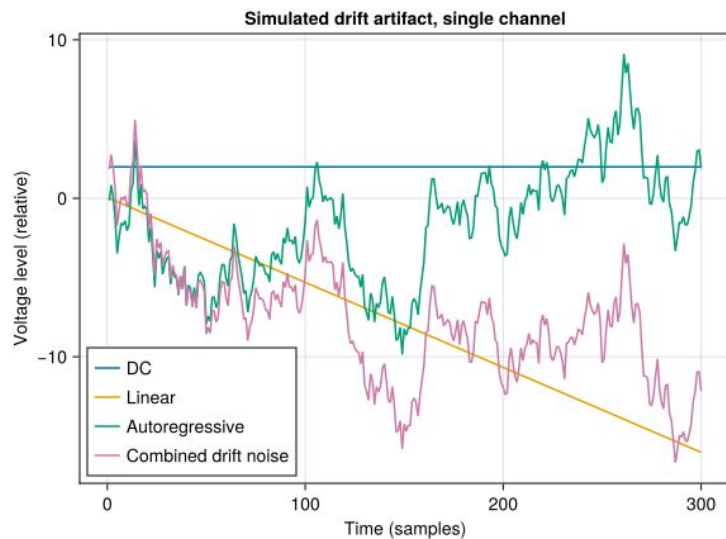
- Choice of weights for individual harmonics
 - Relative to each other
 - Relative to other artifacts



Simulated power line noise, 50Hz + 150Hz

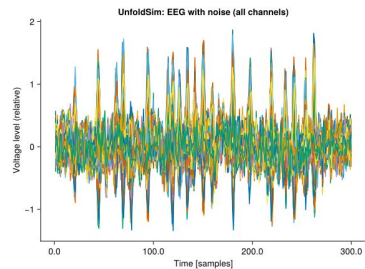
Results: Drift

```
UnfoldSim.DCDriftNoise(scaling_factor=20);  
UnfoldSim.LinearDriftNoise(scaling_factor=20); UnfoldSim.ARDriftNoise()
```

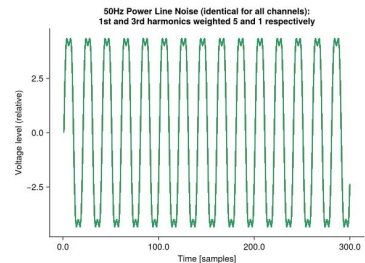


Drift artifact on all channels visualized together

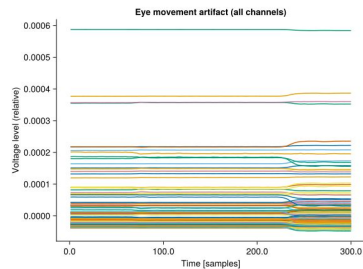
Results - combined



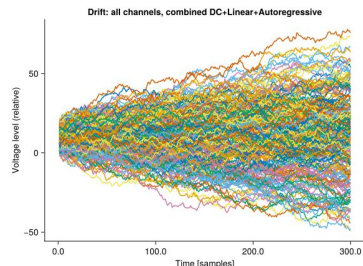
UnfoldSim: EEG + noise



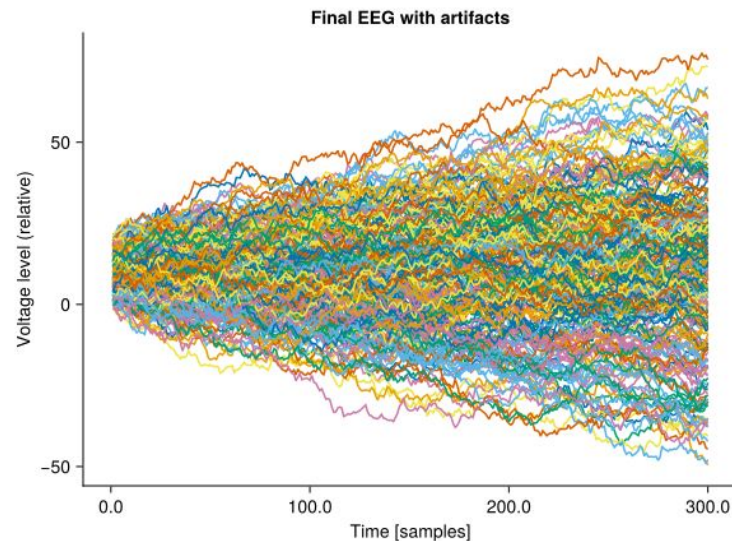
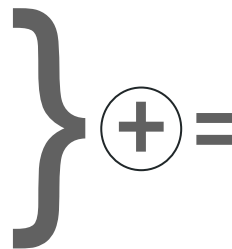
Power line noise



Eye movement

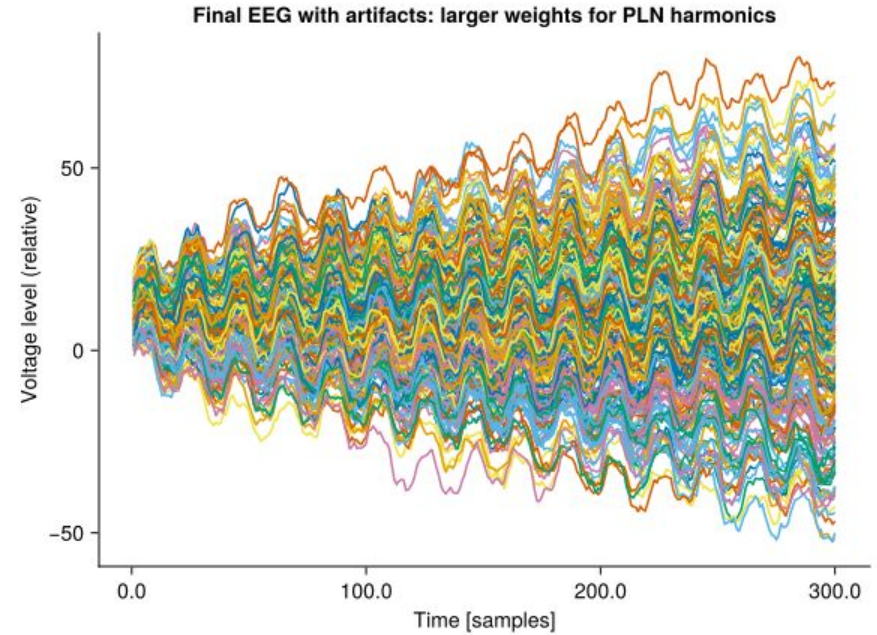
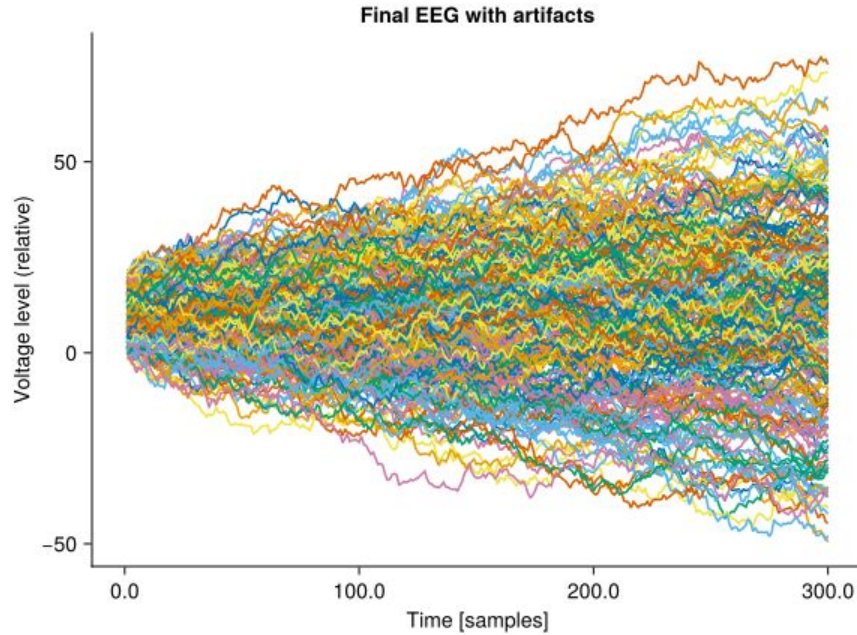


Drift



Simulated EEG with noise and artifacts

Observation: individual artifacts may dominate



Limitations

- User pre-calculates weights
 - Artifact strength with respect to EEG & other artifacts
 - Relative weighting of different parts of the artifact (e.g. harmonics)
- Onsets: by default all artifacts start from time $t=0$
 - User must pre-pad control signal & specify for all time points
- Eye movement:
 - only eyeball modelled (not eyelid)
 - needs more user interfaces

Limitations

- Power line noise:
 - Default: uniform across all channels → realistic?
 - Harmonics weights constant over time
- Drifts:
 - Other models may be more suitable
 - Relative magnitude of sub-parts needs adjustment from user

Outlook

- Artifact scaling
 - Data-driven: calculate relative weights from analyzing real data
 - Recreate artifacts from real data by simulation & compare
- User-defined custom artifact
 - E.g. Artifact extracted from real data, add to UnfoldSim's simulation flow
- User interfaces: more high-level options to control/specify the signal.
 - E.g. provide eye-gaze fixations and onsets, simulator calculates intermediate trajectories

Recap

- This thesis: Base for eye movement, power line noise, drift simulation
- Compatible/integrated with UnfoldSim.jl code structure
- Flexible
 - Extend/add models
 - Add further artifacts
 - User interfaces

 Questions?

References

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